Modeling and Verification of Distributed Systems in TLA+

CS4720 - Research in Program Analysis

2026/Q4

Project description

In this project, you will produce a TLA+ model for a distributed system and analyze the quality of the model using the SysMoBench benchmark and its quality metrics.

The project has two variants, which you can choose based on your research interests:

A Designing the TLA+ specification of the model yourself – in this version, your project will focus more on the design challenges, the relation between the model and implementation, and your abstractions. For this version, we suggest modeling a system with a relatively small codebase.

B Designing the TLA+ specification using GenAI – in this version, your project will focus more on the prompting and analysis of the challenges of model generation using LLM models. For this version, you can work on a production implementation of a distributed system.

System under analysis for your project:

You can test a distributed database system of your choice, such as Cassandra, CockroachDB, etcd, MongoDB, NebulaGraph, Redis, or Riak. Some examples are listed below:

A Chord (sample-1, sample-2), PBFT protocol, FaB protocol

B Essential Paxos, (Hashicorp Raft), Apache Ratis Raft, SmartBFT, hyperledger-labs/SmartBFT.

If you are interested in another system not listed above, contact the lecturers with your proposal.

Project Phases

Phase 1: Protocol Analysis (Week 1, May 12 – May 19)

Read the paper thoroughly and identify:

- The objective of the project, proposed technique, and main results.
- Design of the studied protocol's steps, messages, and event handlers.
- Feasibility assessment (e.g., you can consider modeling a specific part of a distributed system) and plan with task division.

If you are interested in modeling part of a system, make sure to reach out to the lecturers to confirm the project's scope.

Phase 2: Formal modeling of the protocol in TLA+ (Weeks 2–3, May 19 – June 2)

Your model should capture:

- Model the global and local states of the protocol execution
- Model the transitions for protocol steps, exchanging messages, and event handlers.
- Simulate some executions of the protocol.

Phase 3: Verification and performance analysis (Weeks 4–5, June 2 – June 16

- Describe the specified correctness properties
- Explore model checking performance of different configurations of protocol execution
- Compare your results to related works and the performance of verifying similar protocols.

The model and the performance results must be **clearly evaluated and honestly discussed**.

Final Report (due June 19)

Submit a report of 4–8 pages (excluding the statement of GenAI use and references), in double-column ACM format, including:

1. **Introduction:** Motivation, studied distributed protocol, and objectives of your project.
2. **Formal modeling:** Explanation of the formal model design (including the prompts and performance of GenAI tools if used).
3. **Correctness specifications and verification results:** The correctness specification used, model checking parameters, performance, and SysMoBench evaluation.
4. **Limitations:** Limitations of the model and assumptions you used in your analysis.
5. **Discussion of related work:** Discuss the pros/cons of the study compared to most related work and the methods discussed in the lectures.
6. **Statement of GenAI use:** Use of GenAI tools must be disclosed and follow the course policy.
7. **References**

The submitted reports should have a link to the source code and artifacts (alternatively, a zipped file of the repo as supplementary material), along with a README containing build and execution instructions.